Program 1

Output

Summary Statistics

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	0.226180	0.607510	0.566438	0.207439	0.094326	0.476790	0.168179	0.204015	0.348958
std	0.198210	0.160666	0.158654	0.161134	0.136222	0.117499	0.141473	0.196004	0.476951
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.058824	0.497487	0.508197	0.000000	0.000000	0.406855	0.070773	0.050000	0.000000
50%	0.176471	0.587940	0.590164	0.232323	0.036052	0.476900	0.125747	0.133333	0.000000
75%	0.352941	0.704774	0.655738	0.323232	0.150414	0.545455	0.234095	0.333333	1.000000
max	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000

Preprocessed Dataset

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	0.352941	0.743719	0.590164	0.353535	0.000000	0.500745	0.234415	0.483333	1.0
1	0.058824	0.427136	0.540984	0.292929	0.000000	0.396423	0.116567	0.166667	0.0
2	0.470588	0.919598	0.524590	0.000000	0.000000	0.347243	0.253629	0.183333	1.0
3	0.058824	0.447236	0.540984	0.232323	0.111111	0.418778	0.038002	0.000000	0.0
4	0.000000	0.688442	0.327869	0.353535	0.198582	0.642325	0.943638	0.200000	1.0

Datasets (First 10)

1	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
2	6	148	72	35	0	33.6	0.627	50	1
3	1	85	66	29	0	26.6	0.351	31	0
4	8	183	64	0	0	23.3	0.672	32	1
5	1	89	66	23	94	28.1	0.167	21	0
6	0	137	40	35	168	43.1	2.288	33	1
7	5	116	74	0	0	25.6	0.201	30	0
8	3	78	50	32	88	31	0.248	26	1
9	10	115	0	0	0	35.3	0.134	29	0
10	2	197	70	45	543	30.5	0.158	53	1

Program 2

Output

```
Mean Squared Error (MSE): 9.442519317084798
R² Score: 0.8243789497964555
     Actual MPG  Predicted MPG
198        33.0      31.554520
396        28.0      30.689817
33         19.0      21.805460
208        13.0      16.988055
93         14.0      12.605638
```

Datasets (First 10)

	mpg	cylinders	displacement	horsepower	weight	acceleration	model-year
2	18	8	307	130	3504	12	70
3	15	8	350	165	3693	11.5	70
4	18	8	318	150	3436	11	70
5	16	8	304	150	3433	12	70
6	17	8	302	140	3449	10.5	70
7	15	8	429	198	4341	10	70
8	14	8	454	220	4354	9	70
9	14	8	440	215	4312	8.5	70
10	14	8	455	225	4425	10	70

Program 3

Dataset and Output

```
First 10 Records:
         id diagnosis  radius_mean  ...  concave points_worst  symmetry_worst  fractal_dimension_worst
0    842302         M        17.99  ...                0.2654          0.4601                  0.11890
1    842517         M        20.57  ...                0.1860          0.2750                  0.08902
2  84300903         M        19.69  ...                0.2430          0.3613                  0.08758
3  84348301         M        11.42  ...                0.2575          0.6638                  0.17300
4  84358402         M        20.29  ...                0.1625          0.2364                  0.07678
5    843786         M        12.45  ...                0.1741          0.3985                  0.12440
6    844359         M        18.25  ...                0.1932          0.3063                  0.08368
7  84458202         M        13.71  ...                0.1556          0.3196                  0.11510
8    844981         M        13.00  ...                0.2060          0.4378                  0.10720
9  84501001         M        12.46  ...                0.2210          0.4366                  0.20750

[10 rows x 32 columns]

Accuracy Score: 0.9736842105263158

Confusion Matrix:
[[70  1]
 [ 2 41]]

Classification Report:
              precision    recall  f1-score   support

           B       0.97      0.99      0.98        71
           M       0.98      0.95      0.96        43

    accuracy                           0.97       114
   macro avg       0.97      0.97      0.97       114
weighted avg       0.97      0.97      0.97       114
```

```
Actual vs Predicted (First 10 records)
  Actual Predicted
0      B         B
1      M         M
2      M         M
3      B         B
4      B         B
5      M         M
6      M         M
7      M         M
8      B         B
9      B         B
```

Program 4

Dataset and Output

First 10 Records

	RI	Na	Mg	Al	Si	K	Ca	Ba	Fe	Type
0	1.52101	13.64	4.49	1.10	71.78	0.06	8.75	0.0	0.00	1
1	1.51761	13.89	3.60	1.36	72.73	0.48	7.83	0.0	0.00	1
2	1.51618	13.53	3.55	1.54	72.99	0.39	7.78	0.0	0.00	1
3	1.51766	13.21	3.69	1.29	72.61	0.57	8.22	0.0	0.00	1
4	1.51742	13.27	3.62	1.24	73.08	0.55	8.07	0.0	0.00	1
5	1.51596	12.79	3.61	1.62	72.97	0.64	8.07	0.0	0.26	1
6	1.51743	13.30	3.60	1.14	73.09	0.58	8.17	0.0	0.00	1
7	1.51756	13.15	3.61	1.05	73.24	0.57	8.24	0.0	0.00	1
8	1.51918	14.04	3.58	1.37	72.08	0.56	8.30	0.0	0.00	1
9	1.51755	13.00	3.60	1.36	72.99	0.57	8.40	0.0	0.11	1

Dataset Shape: (214, 10)

Column Names:

```
Index(['RI', 'Na', 'Mg', 'Al', 'Si', 'K', 'Ca', 'Ba', 'Fe', 'Type'], dtype='str')
```

Accuracy for Different K Values

```
K = 1  Accuracy = 0.6977
K = 2  Accuracy = 0.7907
K = 3  Accuracy = 0.7907
K = 4  Accuracy = 0.7209
K = 5  Accuracy = 0.7209
K = 6  Accuracy = 0.6977
K = 7  Accuracy = 0.6977
K = 8  Accuracy = 0.7209
K = 9  Accuracy = 0.7209
K = 10  Accuracy = 0.7442
K = 11  Accuracy = 0.6744
K = 12  Accuracy = 0.6512
K = 13  Accuracy = 0.6279
K = 14  Accuracy = 0.6047
K = 15  Accuracy = 0.6744
```

Actual vs Predicted

	Actual	Predicted
0	7	7
1	5	5
2	2	2
3	5	5
4	1	1
5	1	2
6	2	2
7	2	2
8	1	1
9	1	1

```
Best K Value = 2

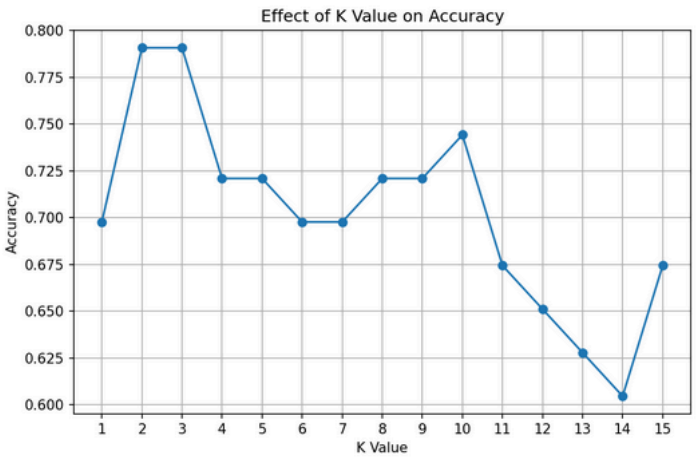
Accuracy Score
0.7906976744186046

Confusion Matrix
[[13  1  0  0  0  0]
 [ 2 12  0  1  0  0]
 [ 2  0  1  0  0  0]
 [ 0  1  0  2  0  0]
 [ 0  1  0  0  1  0]
 [ 0  1  0  0  0  5]]

Classification Report
              precision    recall  f1-score   support

           1       0.76      0.93      0.84        14
           2       0.75      0.80      0.77        15
           3       1.00      0.33      0.50         3
           5       0.67      0.67      0.67         3
           6       1.00      0.50      0.67         2
           7       1.00      0.83      0.91         6

    accuracy                           0.79        43
   macro avg       0.86      0.68      0.73        43
weighted avg       0.81      0.79      0.78        43
```



Program 5

Dataset and Output

```
First 10 Records:
   age           job   marital  education default  balance housing loan  contact  day month  duration  campaign  pdays  previous poutcome Target
0   58    management   married   tertiary      no     2143     yes   no  unknown    5   may       261         1     -1         0  unknown     no
1   44    technician    single  secondary      no       29     yes   no  unknown    5   may       151         1     -1         0  unknown     no
2   33  entrepreneur   married  secondary      no        2     yes  yes  unknown    5   may        76         1     -1         0  unknown     no
3   47   blue-collar   married    unknown      no     1506     yes   no  unknown    5   may        92         1     -1         0  unknown     no
4   33       unknown    single    unknown      no        1      no   no  unknown    5   may       198         1     -1         0  unknown     no
5   35    management   married   tertiary      no      231     yes   no  unknown    5   may       139         1     -1         0  unknown     no
6   28    management    single   tertiary      no      447     yes  yes  unknown    5   may       217         1     -1         0  unknown     no
7   42  entrepreneur  divorced   tertiary     yes        2     yes   no  unknown    5   may       380         1     -1         0  unknown     no
8   58       retired   married    primary      no      121     yes   no  unknown    5   may        50         1     -1         0  unknown     no
9   43    technician    single  secondary      no      593     yes   no  unknown    5   may        55         1     -1         0  unknown     no

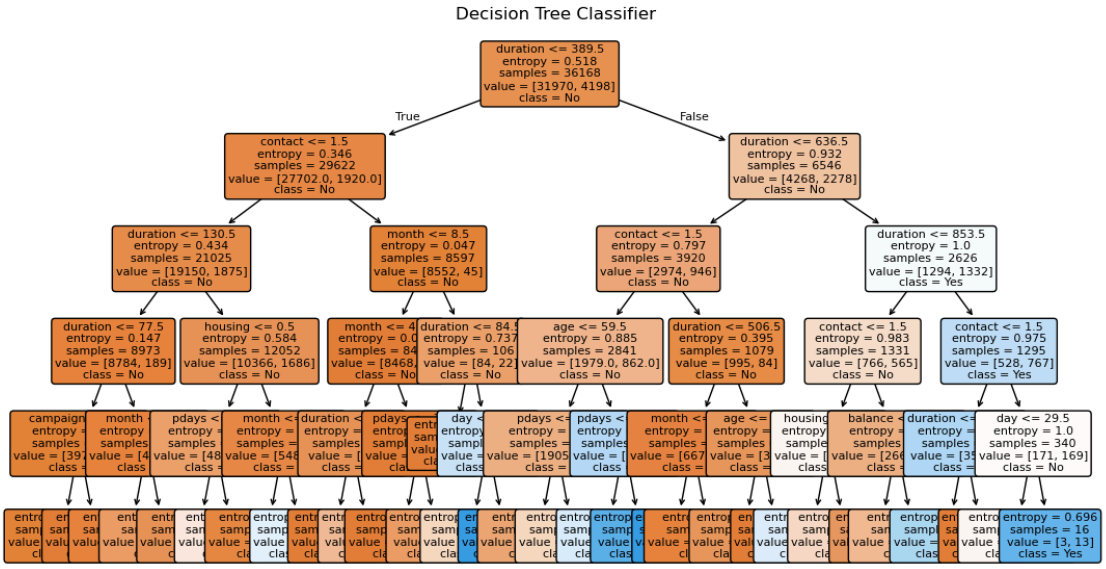
Accuracy: 0.8848833351763795
Precision: 0.5548245614035088

Confusion Matrix
[[7749  203]
 [ 838  253]]

Classification Report
              precision    recall  f1-score   support

           0       0.90      0.97      0.94      7952
           1       0.55      0.23      0.33      1091

    accuracy                           0.88      9043
   macro avg       0.73      0.60      0.63      9043
weighted avg       0.86      0.88      0.86      9043
```



Program 6

Output

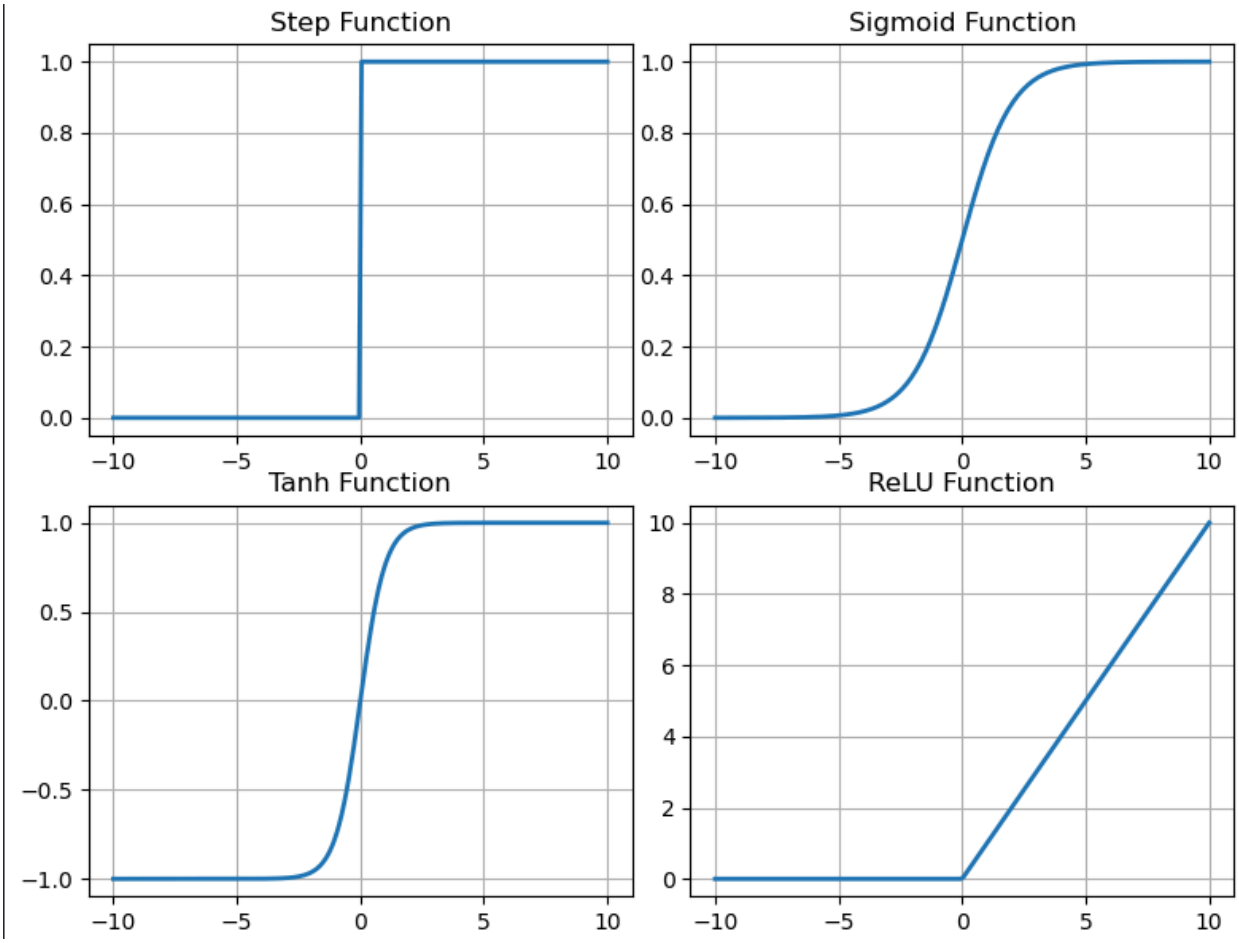
```
Weights: [[2. 2.]]
Bias: [-3.]
OR Gate using Perceptron

Input   Expected        Predicted
[0 0]   0               0
[0 1]   1               1
[1 0]   1               1
[1 1]   1               1

Accuracy: 1.0
Weights: [[2. 2.]]
Bias: [-1.]
```

Program 7

Output



Program 8

Output

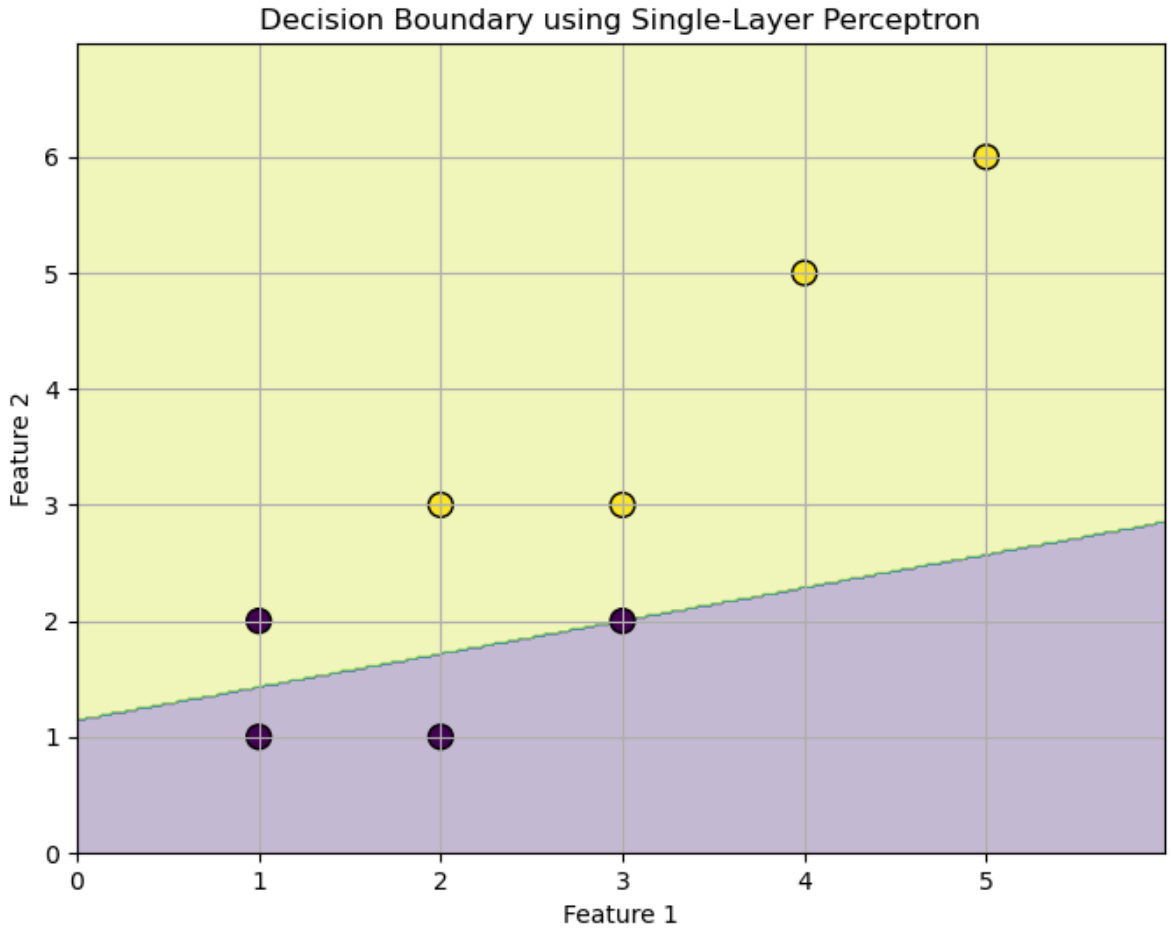
```
Input   Expected        Predicted
[0 0]   0               0
[0 1]   1               0
[1 0]   1               0
[1 1]   0               0

Accuracy: 0.5
Weights: [[0. 0.]]
Bias: [0.]
```

Program 9

Output

Training Accuracy: 0.875



Program 10

Output

```
 [ 0  0 11]]

Classification Report
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      0.89      0.94         9
           2       0.92      1.00      0.96        11

    accuracy                           0.97        30
   macro avg       0.97      0.96      0.97        30
weighted avg       0.97      0.97      0.97        30
```

Program 11

Output

```
 [ 0  0 11]]

Classification Report
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         9
           2       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

Program 12

Output

```
SGD Optimizer
Accuracy: 1.0

SGD with Momentum
Accuracy: 1.0
```

Program 13

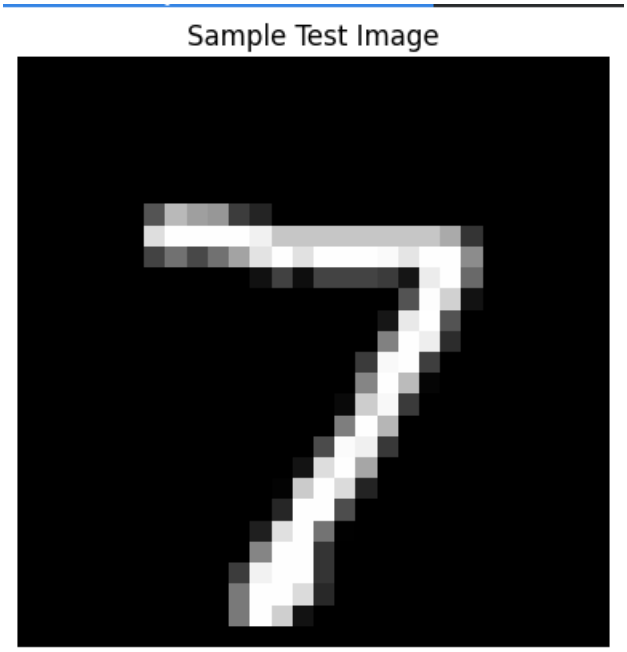
Output

Hidden Neurons	Training Error	Testing Error
2	0.0333	0.0333
4	0.0167	0.0000
6	0.0333	0.0333
8	0.0333	0.0000
10	0.0250	0.0000
15	0.0250	0.0333
20	0.0250	0.0000
30	0.0333	0.0000

Program 14

Output

Test Accuracy: 0.986299991607666



Predicted Digit: 7